

Shreyas Thakur

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Thane, Maharashtra, India – 400601

PROFESSIONAL SUMMARY

Chemical Engineering graduate from the Institute of Chemical Technology, Mumbai (2025) with hands-on industrial experience in continuous process operations and troubleshooting at Epigral Ltd. Skilled in process simulation, HAZOP, PFD/P&ID development, and equipment design. Seeking a Process Engineer role to apply technical expertise in plant optimization, scale-up, and safety excellence.

TECHNICAL SKILLS

Process Engineering: Mass & Energy Balances, Unit Operations, Reaction Engineering, Heat/Mass Transfer, Thermodynamics

Safety & Compliance: HAZOP Analysis, Process Safety Management (PSM), Management of Change (MOC), Effluent Treatment

Design & Software: Aspen HYSYS, MATLAB, Python, AutoCAD, PFD, P&ID, Equipment Sizing, Advanced MS Excel, Power BI

Operations: DCS (Distributed Control System), Continuous & Batch Processing, Root Cause Analysis, SOP Development

EDUCATION

Institute of Chemical Technology (ICT), Mumbai

Mumbai, India

Bachelor of Technology – Chemical Engineering | CGPA: 6.6

2021 – 2025

- **Coursework:** Reaction Eng, Heat & Mass Transfer, Process Control, Thermodynamics, Plant Design
- HSC: 85.67% | SSC: 85.20%

EXPERIENCE

In-Plant Trainee – Process Engineering

Jun 2024 – Aug 2024

Epigral Ltd. (Formerly Meghmani Finechem)

Dahej, Gujarat, India

- Performed **root cause analysis** on catalyst deactivation in Hydrogen Peroxide unit, improving lifecycle monitoring.
- Investigated recurring **burner failures** in HCl synthesis, proposing corrective actions aligned with **PSM protocols**.
- Conducted a **moisture-reduction study** in Epichlorohydrin (ECH) purification to identify and resolve bottlenecks.
- Operated continuous chemical processes via **DCS**, gaining real-time exposure to process parameters and alarms.
- Completed training in industrial safety, HAZOP procedures, permit-to-work systems, and **MOC documentation**.

PROJECTS

Plant Design – Ambrettolide Manufacturing (25 TPA) | Capstone

Sep 2024 – Apr 2025

- Designed an end-to-end chemical plant, evaluating 3 synthesis routes via techno-economic and safety analysis.
- Performed rigorous **material/energy balances** and sized equipment (CSTR, distillation columns, heat exchangers).
- Developed comprehensive **PFD/P&ID** and conducted detailed **HAZOP** identifying 15+ deviation scenarios.
- Prepared complete project cost estimation (**CAPEX/OPEX**), ROI analysis, site layout, and effluent treatment scheme.

Impact of Climate Change on Water Quality | Seminar

Nov 2024 – Jan 2025

- Authored a technical review analyzing climate change impacts on water systems in reference to **CPCB standards**.
- Benchmarked advanced water treatment technologies (**UF/NF/RO, AOPs**) for industrial and municipal scalability.

CERTIFICATIONS & ACHIEVEMENTS

Certifications: Petroleum Refinery Engineering (IChE) | Industrial Environmental Pollution Mgmt (Terra-Green) | Industrial IoT (NPTEL - Top 5% Nationwide)

Leadership: **Creative Head**, *Bombay Technologist* (ICT Mumbai's Official Technical Magazine, 2023-2024)

Competitions: **2nd Place**, "The Entrepreneur's Blueprint" business plan competition at Vortex 2022, ICT Mumbai